

Research Assignment - Module 01

The Language of Business Analytics

Section 1: Foundational Concepts of Analytics

1 Business Analytics: When companies use data to understand what's going on in the business and make better decisions. It's not just looking at numbers, but actually using them to improve performance and strategy.

Example: Shoprite uses sales data to decide which products to stock more of in different areas.

Source: Investopedia (2024)

2 Data Analytics: The process of analysing raw data to find useful insights. It focuses more on the data itself, while business analytics focuses on decision-making.

Example: Looking at how many people buy airline on different days to see when sales peak.

Source: IBM ⁽²⁰²⁴⁾ ~~Data Analytics Guide~~

3. Business Intelligence: Tools and systems that collect and present data in an easy way, like dashboards or reports. It mainly helps track what is happening in the business right now or in the past.

Example: A dashboard showing daily sales across Pick n Pay stores.

Source: Microsoft BI Documentation

4. Data - Driven Decision Making: When decisions are made based on data instead of just gut feel or assumptions it helps businesses reduce risk and be more accurate.

Example: A company deciding to open a new store in Soweto after seeing high customer demand data from that area.

Source: Harvard Business Review

5. Big Data: Big Data refers to very large amounts of data that come in quickly and from different sources. It is often too complex to handle using normal tools.

Example: Data collected from millions of users on apps like WhatsApp or TikTok everyday.

Source: Oracle (2024)

6. Descriptive Analytics: Looking at past data to explain what has already happened.

Example: A ~~per~~ report showing how much revenue a company made last month.

Source: McKinsey & Company, 2024

7. Diagnostic Analytics: This type of analytics tries to explain why something happened by digging deeper into the data.

Example: Finding out that sales dropped because of load shedding affecting store operations.

Source: IBM (2024)

8. Predictive Analytics: Uses past data to make predictions about the future. It often uses patterns and models.

Example: Banks predicting which customers are likely to default on loans.

Source: SAS Institute (2024)

9. Prescriptive Analytics: This type of analytics suggests what actions a business should take based on the data. It goes beyond prediction to give recommendations.

Example: A delivery company suggesting the best routes to avoid traffic in ~~Joh~~ Johannesburg CBD.

Source: Deloitte (2024)

10. Data Literacy: The ability to read, understand and use data correctly. ~~It means~~

Example: A store manager being able to understand a sales report and make decisions from it.

Source: Glatner, 2025

Section 2: Data, Measurement and Reporting

11. Structured Data: Data that is organised in rows and columns, like in Excel or databases. It is easy to store and analyse.

Example: Customer list with names, ages and phone numbers

Source: IBM (2024)

12. Unstructured data: Data that is not organised in a fixed format, like text, videos or images. It is much harder to analyse.

Example: Comments on a X post about a brand.

Source: Oracle

13. Data Warehouse: A central place where a company stores large amounts of data for analysis and reporting.

Example: A bank storing all transactions data in one system for reporting.

Source: Amazon AWS

14. Data Mining: The process of finding patterns or trends in large sets of data.

Example: Discovering that customers who buy bread also often buy milk.

Source: IBM (2024)

15. Data Visualisation: Presenting data using charts, graphs or visuals to make it easier to understand

Example: A bar chart showing monthly sales

Source: Tableau

16. Dashboard: A visual display that shows important data and KPIs in one place, usually updated in real time

Example: A sales dashboard showing daily performance of stores.

Source: Microsoft

17. Metric: Any number used to measure something in a business

Example: Number of website visits

Source: HubSpot

18. Key Performance Indicator (KPI): A metric that is important for tracking business success and guiding decisions

Example: Monthly revenue target

Source: Investopedia.